

Headers

The diagrams illustrate the pin connections for four different communication headers on a microcontroller:

- I2C:** A 5-pin header with pins 1, 2, 3, 4, and 5. Pin 1 is connected to GND. Pin 2 is connected to +3.3V. Pin 3 is connected to SDA. Pin 4 is connected to SCL. Pin 5 is connected to GND.
- SPI2:** A 5-pin header with pins 1, 2, 3, 4, and 5. Pin 1 is connected to GND. Pin 2 is connected to +3.3V. Pin 3 is connected to SPI2_CS. Pin 4 is connected to SPI2_MOSI. Pin 5 is connected to SPI2_MISO.
- SWD:** A 5-pin header with pins 1, 2, 3, 4, and 5. Pin 1 is connected to GND. Pin 2 is connected to SWDIO. Pin 3 is connected to SWCLK. Pin 4 is connected to GND. Pin 5 is connected to GND.
- UART5:** A 5-pin header with pins 1, 2, 3, 4, and 5. Pin 1 is connected to GND. Pin 2 is connected to +3.3V. Pin 3 is connected to UART5_TX. Pin 4 is connected to UART5_RX. Pin 5 is connected to GND.

[illegible]

ETHERNET

The schematic diagram illustrates the Ethernet interface circuit for the LAN7720A. The central component is the LAN7720A Ethernet controller, which is connected to the PHY80E1 Ethernet PHY. The LAN7720A is also connected to the J001D01BNL LED indicator. The circuit is powered by a +3.3V supply and ground (GND).

Key Components and Connections:

- LAN7720A:** The main Ethernet controller. It has pins for ETH_TXD0, ETH_TXD1, ETH_TXD2, ETH_TXD3, ETH_TXD4, ETH_TXD5, ETH_TXD6, ETH_TXD7, ETH_TXD8, ETH_TXD9, ETH_TXD10, ETH_TXD11, ETH_TXD12, ETH_TXD13, ETH_TXD14, ETH_TXD15, ETH_TXD16, ETH_TXD17, ETH_TXD18, ETH_TXD19, ETH_TXD20, ETH_TXD21, ETH_TXD22, ETH_TXD23, ETH_TXD24, ETH_TXD25, ETH_TXD26, ETH_TXD27, ETH_TXD28, ETH_TXD29, ETH_TXD30, ETH_TXD31, ETH_TXD32, ETH_TXD33, ETH_TXD34, ETH_TXD35, ETH_TXD36, ETH_TXD37, ETH_TXD38, ETH_TXD39, ETH_TXD40, ETH_TXD41, ETH_TXD42, ETH_TXD43, ETH_TXD44, ETH_TXD45, ETH_TXD46, ETH_TXD47, ETH_TXD48, ETH_TXD49, ETH_TXD50, ETH_TXD51, ETH_TXD52, ETH_TXD53, ETH_TXD54, ETH_TXD55, ETH_TXD56, ETH_TXD57, ETH_TXD58, ETH_TXD59, ETH_TXD60, ETH_TXD61, ETH_TXD62, ETH_TXD63, ETH_TXD64, ETH_TXD65, ETH_TXD66, ETH_TXD67, ETH_TXD68, ETH_TXD69, ETH_TXD70, ETH_TXD71, ETH_TXD72, ETH_TXD73, ETH_TXD74, ETH_TXD75, ETH_TXD76, ETH_TXD77, ETH_TXD78, ETH_TXD79, ETH_TXD80, ETH_TXD81, ETH_TXD82, ETH_TXD83, ETH_TXD84, ETH_TXD85, ETH_TXD86, ETH_TXD87, ETH_TXD88, ETH_TXD89, ETH_TXD90, ETH_TXD91, ETH_TXD92, ETH_TXD93, ETH_TXD94, ETH_TXD95, ETH_TXD96, ETH_TXD97, ETH_TXD98, ETH_TXD99, ETH_TXD100, ETH_TXD101, ETH_TXD102, ETH_TXD103, ETH_TXD104, ETH_TXD105, ETH_TXD106, ETH_TXD107, ETH_TXD108, ETH_TXD109, ETH_TXD110, ETH_TXD111, ETH_TXD112, ETH_TXD113, ETH_TXD114, ETH_TXD115, ETH_TXD116, ETH_TXD117, ETH_TXD118, ETH_TXD119, ETH_TXD120, ETH_TXD121, ETH_TXD122, ETH_TXD123, ETH_TXD124, ETH_TXD125, ETH_TXD126, ETH_TXD127, ETH_TXD128, ETH_TXD129, ETH_TXD130, ETH_TXD131, ETH_TXD132, ETH_TXD133, ETH_TXD134, ETH_TXD135, ETH_TXD136, ETH_TXD137, ETH_TXD138, ETH_TXD139, ETH_TXD140, ETH_TXD141, ETH_TXD142, ETH_TXD143, ETH_TXD144, ETH_TXD145, ETH_TXD146, ETH_TXD147, ETH_TXD148, ETH_TXD149, ETH_TXD150, ETH_TXD151, ETH_TXD152, ETH_TXD153, ETH_TXD154, ETH_TXD155, ETH_TXD156, ETH_TXD157, ETH_TXD158, ETH_TXD159, ETH_TXD160, ETH_TXD161, ETH_TXD162, ETH_TXD163, ETH_TXD164, ETH_TXD165, ETH_TXD166, ETH_TXD167, ETH_TXD168, ETH_TXD169, ETH_TXD170, ETH_TXD171, ETH_TXD172, ETH_TXD173, ETH_TXD174, ETH_TXD175, ETH_TXD176, ETH_TXD177, ETH_TXD178, ETH_TXD179, ETH_TXD180, ETH_TXD181, ETH_TXD182, ETH_TXD183, ETH_TXD184, ETH_TXD185, ETH_TXD186, ETH_TXD187, ETH_TXD188, ETH_TXD189, ETH_TXD190, ETH_TXD191, ETH_TXD192, ETH_TXD193, ETH_TXD194, ETH_TXD195, ETH_TXD196, ETH_TXD197, ETH_TXD198, ETH_TXD199, ETH_TXD200, ETH_TXD201, ETH_TXD202, ETH_TXD203, ETH_TXD204, ETH_TXD205, ETH_TXD206, ETH_TXD207, ETH_TXD208, ETH_TXD209, ETH_TXD210, ETH_TXD211, ETH_TXD212, ETH_TXD213, ETH_TXD214, ETH_TXD215, ETH_TXD216, ETH_TXD217, ETH_TXD218, ETH_TXD219, ETH_TXD220, ETH_TXD221, ETH_TXD222, ETH_TXD223, ETH_TXD224, ETH_TXD225, ETH_TXD226, ETH_TXD227, ETH_TXD228, ETH_TXD229, ETH_TXD230, ETH_TXD231, ETH_TXD232, ETH_TXD233, ETH_TXD234, ETH_TXD235, ETH_TXD236, ETH_TXD237, ETH_TXD238, ETH_TXD239, ETH_TXD240, ETH_TXD241, ETH_TXD242, ETH_TXD243, ETH_TXD244, ETH_TXD245, ETH_TXD246, ETH_TXD247, ETH_TXD248, ETH_TXD249, ETH_TXD250, ETH_TXD251, ETH_TXD252, ETH_TXD253, ETH_TXD254, ETH_TXD255, ETH_TXD256, ETH_TXD257, ETH_TXD258, ETH_TXD259, ETH_TXD260, ETH_TXD261, ETH_TXD262, ETH_TXD263, ETH_TXD264, ETH_TXD265, ETH_TXD266, ETH_TXD267, ETH_TXD268, ETH_TXD269, ETH_TXD270, ETH_TXD271, ETH_TXD272, ETH_TXD273, ETH_TXD274, ETH_TXD275, ETH_TXD276, ETH_TXD277, ETH_TXD278, ETH_TXD279, ETH_TXD280, ETH_TXD281, ETH_TXD282, ETH_TXD283, ETH_TXD284, ETH_TXD285, ETH_TXD286, ETH_TXD287, ETH_TXD288, ETH_TXD289, ETH_TXD290, ETH_TXD291, ETH_TXD292, ETH_TXD293, ETH_TXD294, ETH_TXD295, ETH_TXD296, ETH_TXD297, ETH_TXD298, ETH_TXD299, ETH_TXD300, ETH_TXD301, ETH_TXD302, ETH_TXD303, ETH_TXD304, ETH_TXD305, ETH_TXD306, ETH_TXD307, ETH_TXD308, ETH_TXD309, ETH_TXD310, ETH_TXD311, ETH_TXD312, ETH_TXD313, ETH_TXD314, ETH_TXD315, ETH_TXD316, ETH_TXD317, ETH_TXD318, ETH_TXD319, ETH_TXD320, ETH_TXD321, ETH_TXD322, ETH_TXD323, ETH_TXD324, ETH_TXD325, ETH_TXD326, ETH_TXD327, ETH_TXD328, ETH_TXD329, ETH_TXD330, ETH_TXD331, ETH_TXD332, ETH_TXD333, ETH_TXD334, ETH_TXD335, ETH_TXD336, ETH_TXD337, ETH_TXD338, ETH_TXD339, ETH_TXD340, ETH_TXD341, ETH_TXD342, ETH_TXD343, ETH_TXD344, ETH_TXD345, ETH_TXD346, ETH_TXD347, ETH_TXD348, ETH_TXD349, ETH_TXD350, ETH_TXD351, ETH_TXD352, ETH_TXD353, ETH_TXD354, ETH_TXD355, ETH_TXD356, ETH_TXD357, ETH_TXD358, ETH_TXD359, ETH_TXD360, ETH_TXD361, ETH_TXD362, ETH_TXD363, ETH_TXD364, ETH_TXD365, ETH_TXD366, ETH_TXD367, ETH_TXD368, ETH_TXD369, ETH_TXD370, ETH_TXD371, ETH_TXD372, ETH_TXD373, ETH_TXD374, ETH_TXD375, ETH_TXD376, ETH_TXD377, ETH_TXD378, ETH_TXD379, ETH_TXD380, ETH_TXD381, ETH_TXD382, ETH_TXD383, ETH_TXD384, ETH_TXD385, ETH_TXD386, ETH_TXD387, ETH_TXD388, ETH_TXD389, ETH_TXD390, ETH_TXD391, ETH_TXD392, ETH_TXD393, ETH_TXD394, ETH_TXD395, ETH_TXD396, ETH_TXD397, ETH_TXD398, ETH_TXD399, ETH_TXD400, ETH_TXD401, ETH_TXD402, ETH_TXD403, ETH_TXD404, ETH_TXD405, ETH_TXD406, ETH_TXD407, ETH_TXD408, ETH_TXD409, ETH_TXD410, ETH_TXD411, ETH_TXD412, ETH_TXD413, ETH_TXD414, ETH_TXD415, ETH_TXD416, ETH_TXD417, ETH_TXD418, ETH_TXD419, ETH_TXD420, ETH_TXD421, ETH_TXD422, ETH_TXD423, ETH_TXD424, ETH_TXD425, ETH_TXD426, ETH_TXD427, ETH_TXD428, ETH_TXD429, ETH_TXD430, ETH_TXD431, ETH_TXD432, ETH_TXD433, ETH_TXD434, ETH_TXD435, ETH_TXD436, ETH_TXD437, ETH_TXD438, ETH_TXD439, ETH_TXD440, ETH_TXD441, ETH_TXD442, ETH_TXD443, ETH_TXD444, ETH_TXD445, ETH_TXD446, ETH_TXD447, ETH_TXD448, ETH_TXD449, ETH_TXD450, ETH_TXD451, ETH_TXD452, ETH_TXD453, ETH_TXD454, ETH_TXD455, ETH_TXD456, ETH_TXD457, ETH_TXD458, ETH_TXD459, ETH_TXD460, ETH_TXD461, ETH_TXD462, ETH_TXD463, ETH_TXD464, ETH_TXD465, ETH_TXD466, ETH_TXD467, ETH_TXD468, ETH_TXD469, ETH_TXD470, ETH_TXD471, ETH_TXD472, ETH_TXD473, ETH_TXD474, ETH_TXD475, ETH_TXD476, ETH_TXD477, ETH_TXD478, ETH_TXD479, ETH_TXD480, ETH_TXD481, ETH_TXD482, ETH_TXD483, ETH_TXD484, ETH_TXD485, ETH_TXD486, ETH_TXD487, ETH_TXD488, ETH_TXD489, ETH_TXD490, ETH_TXD491, ETH_TXD492, ETH_TXD493, ETH_TXD494, ETH_TXD495, ETH_TXD496, ETH_TXD497, ETH_TXD498, ETH_TXD499, ETH_TXD500, ETH_TXD501, ETH_TXD502, ETH_TXD503, ETH_TXD504, ETH_TXD505, ETH_TXD506, ETH_TXD507, ETH_TXD508, ETH_TXD509, ETH_TXD510, ETH_TXD511, ETH_TXD512, ETH_TXD513, ETH_TXD514, ETH_TXD515, ETH_TXD516, ETH_TXD517, ETH_TXD518, ETH_TXD519, ETH_TXD520, ETH_TXD521, ETH_TXD522, ETH_TXD523, ETH_TXD524, ETH_TXD525, ETH_TXD526, ETH_TXD527, ETH_TXD528, ETH_TXD529, ETH_TXD530, ETH_TXD531, ETH_TXD532, ETH_TXD533, ETH_TXD534, ETH_TXD535, ETH_TXD536, ETH_TXD537, ETH_TXD538, ETH_TXD539, ETH_TXD540, ETH_TXD541, ETH_TXD542, ETH_TXD543, ETH_TXD544, ETH_TXD545, ETH_TXD546, ETH_TXD547, ETH_TXD548, ETH_TXD549, ETH_TXD550, ETH_TXD551, ETH_TXD552, ETH_TXD553, ETH_TXD554,